Thermal Properties of Matter

Single Correct Type Questions

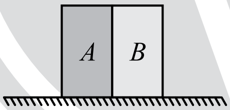
1. At what temperature a gold ring of diameter 6.230 cm be heated so that it can be fitted on a wooden bangle of diameter. 6.241 cm? Both the diameters have been measured at room temperature (27°C).

(Given: coefficient of linear thermal expansion of gold $\alpha_L = 1.4 \times 10^{-5}\ K^{-1}$) **[29 June, 2022 (Shift-II)]**

(*a*) 125.7°C (*b*) 91.7°C
(*c*) 425.7°C (*d*) 152.7°C

2. A bimetallic strip consists of metals A and B. It is mounted rigidly as shown. The metal A has higher coefficient of expansion compared to that metal B. When bimetallic strip is placed in a cold bath, it will

[16 March, 2021 (Shift-II)]



(*a*) Neither bend nor shrink
(*b*) Not bend but shrink
(*c*) Bend towards the left
(*d*) Bend towards the right

3. Two different wires having lengths L_1 and L_2, and respective temperature coefficient of linear expansion α_1 and α_2, are joined end-to-end. Then the effective temperature coefficient of linear expansion is

[5 Sep, 2020 (Shift-II)]

(*a*) $\sqrt[2]{\alpha_1\alpha_2}$ (*b*) $\dfrac{4\alpha_1\alpha_2}{\alpha_1+\alpha_2}\dfrac{L_2L_1}{(L_2+L_1)^2}$

(*c*) $\dfrac{\alpha_1+\alpha_2}{2}$ (*d*) $\dfrac{\alpha_1L_1+\alpha_2L_2}{L_1+L_2}$

4. Two rods A and B of identical dimensions are at temperature 30°C. If a heated upto 180°C and *B* upto *T*°C, then the new lengths are the same. If the ratio of the coefficients of linear expansion of *A* and *B* is 4 : 3, then the value of *T* is: **[11 Jan, 2019 (Shift-II)]**

(*a*) 230°C (*b*) 270°C
(*c*) 200°C (*d*) 250°C

5. Heat energy of 184 kJ is given to a block of ice of mass 600*g* at –12°C, Specific heat of ice is $2222.3\,J\,kg^{-1}{}^{\circ}C^{-1}$ and latent heat of ice in 336 kJ/kg^{-1}

[29 Jan, 2023 (Shift-II)]

A. Final temperature of the system will be 0°C.
B. Final temperature of the system will be greater than 0°C.
C. The final system will have a mixture of ice and water in the ratio of 5 : 1.
D. The final system will have a mixture of ice and water in the ratio of 1 : 5.
E. The final system will have water only.

Choose the correct answer from the options given below:

(*a*) A and D only (*b*) B and D only
(*b*) A and E only (*d*) A and C only

6. A copper block of mass 5.0 kg is heated to a temperature of 500°C and is placed on a large ice block. What is the maximum amount of ice that can melt? [Specific heat of copper: 0.39 J g^{-1} °C^{-1} and latent heat of fusion of water: 335 J g^{-1} °C^{-1} **[25 June, 2022 (Shift-II)]**

(*a*) 1.5 kg (*b*) 5.8 kg
(*c*) 2.9 kg (*d*) 3.8 kg

7. The temperature of equal masses of three different liquids *x*, *y* and *z* are 10°C, 20°C and 30°C respectively. The temperature of mixture when *x* is mixed with *y* is 16°C and that when *y* is mixed with *z* is 26°C. The temperature of mixture when *x* and *z* are mixed will be:

[26 Aug, 2021 (Shift-II)]

(*a*) 28.32°C (*b*) 23.84°C
(*c*) 25.62°C (*d*) 20.28°C

8. The height of victoria falls is 63 m. What is the difference in temperature of water at the top and at the bottom of fall?

[Given 1 cal = 4.2 J and specific heat of water = 1 cal g^{-1} °C^{-1}]

[27 Aug, 2021 (Shift-II)]

(*a*) 0.147 ºC (*b*) 14.76 ºC

(*c*) 1.476 ºC (*d*) 0.014 ºC

9. Due to cold weather a 1 m water pipe of cross–sectional area 1 cm^2 is filled with ice at –10°C. Resistive heating is used to melt the ice. Current of 0.5 A is passed through 4 k Ω resistance. Assuming that all the heat produced is used for melting, what is the minimum time required?

(Given latent heat of fusion for water/ice = 3.33×10^5 J kg^{-1}, specific heat of ice = 2×10^3 J kg^{-1} and density of ice = 10^3 kg / m^3) **[1 Sep, 2021 (Shift-II)]**

(*a*) 3.53 s (*b*) 0.353 s

(*c*) 35.3 s (*d*) 70.6 s

10. A bullet of mass 5g, traveling with a speed of 210 m/s, strikes a fixed wooden target. One half of its kinetic energy is converted into heat in the bullet while the other half is converted into heat in the wood. The rise of temperature of the bullet if the specific heat of its material is 0.030 cal/(*g* – ° C) (1 cal = 4.2×10^7 ergs) close to: **[5 Sep, 2020 (Shift-I)]**

(*a*) 83.3°C (*b*) 38.4°C

(*c*) 87.5°C (*d*) 119.2°C

11. A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in °C, if this thermometer in the contact with the object reads $x_0/2$? **[11 Jan, 2019 (Shift-II)]**

(*a*) 25 (*b*) 60 (*c*) 40 (*d*) 45

12. Ice at – 20°C is added to 50 g of water at 40°C. When the temperature of the mixture reaches 0°C, it is found that 20 g of ice is still unmelted. The amount of ice added to the water was close to

(Specific heat of water = 4.2 J/g/°C, Heat of fusion of water at 0°C = 334 J/g and Specific heat of ice = 2.1 J/g/°C)

[11 Jan, 2019 (Shift-I)]

(*a*) 50 g (*b*) 100 g

(*c*) 60 g (*d*) 40 g

13. When 100g of a liquid *A* at 100°C is added to 50g of a liquid *B* at temperature 75°C, the temperature of the mixture becomes 90°C. The temperature of the mixture, if 100g of liquid *A* at 100°C is added to 50g of liquid *B* at 50°C, will be: **[11 Jan, 2019 (Shift-II)]**

(*a*) 85°C (*b*) 60°C

(*c*) 80°C (*d*) 70°C

14. A metal ball of mass 0.1 kg is heated upto 500ºC and dropped into a vessel of heat capacity 800 JK^{-1} and containing 0.5 kg water. The initial temperature of water and vessel is 30ºC. What is the approximate percentage increment in the temperature of the water? [Specific heat Capacitites of water and metal are, respectively 4200 $Jkg^{-1}K^{-1}$ and 400 $Jkg^{-1}K^{-1}$]

[11 Jan, 2019 (Shift-II)]

(*a*) 15% (*b*) 30%

(*c*) 25% (*d*) 20%

15. An ice cube of dimensions 60cm × 50cm × 20cm is placed in an insulation box of wall thickness 1cm. The box keeping the ice cube at 0ºC of temperature is brought to a room of temperature 40ºC. The rate of melting of ice is approximately: **[26 July, 2022 (Shift-II)]**

(Latent heat of fusion of ice is 3.4×10^5 J kg^{-1} and thermal conducting of insulation wall is 0.05 Wm^{-1}ºC^{-1})

(*a*) 61×10^{-3} kg s^{-1} (*b*) 61×10^{-5} kg s^{-1}

(*c*) 208 kg s^{-1} (*d*) 30×10^{-3} kg s^{-1}

16. Two thin metallic spherical shells of radii r_1 and $r_2 (r_1 < r_2)$ are placed with their centres coinciding. A material of thermal conductivity *K* is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature $\theta_2 (\theta_1 < \theta_2)$. The rate at which heat flows radially through the material is:

[31 Aug, 2021 (Shift-II)]

(*a*) $\dfrac{\pi r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$ (*b*) $\dfrac{K(\theta_2 - \theta_1)(r_2 - r_1)}{4\pi r_1 r_2}$

(*c*) $\dfrac{K(\theta_2 - \theta_1)}{r_2 - r_1}$ (*d*) $\dfrac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$

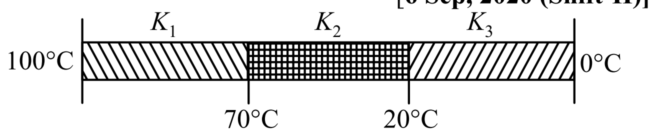
17. Two identical metal of thermal conductivities K_1 and K_2 respectively are connected in series. The effective thermal conductivity of the combination is:

[17 March, 2021 (Shift-I)]

(*a*) $\dfrac{K_1 + K_2}{K_1 K_2}$ (*b*) $\dfrac{K_1 + K_2}{2 K_1 K_2}$ (*c*) $\dfrac{2 K_1 K_2}{K_1 + K_2}$ (*d*) $\dfrac{K_1 K_2}{K_1 + K_2}$

18. Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity K_1, K_2 and K_3, respectively. They are joined together at their ends to make a long rod (see figure). One end of the long rod is maintained at 100°C and the other at 0°C (see figure). If the joints of the rod are at 70°C and 20°C in steady state and there is no loss of energy from the surface of the rod, the correct relationship between K_1, K_2 and K_3 is

[6 Sep, 2020 (Shift-II)]



(*a*) $K_1 : K_3 = 2 : 3$; $K_2 : K_3 = 2 : 5$

(*b*) $K_1 < K_2 < K_3$

(*c*) $K_1 : K_2 = 5 : 2$; $K_1 : K_3 = 3 : 5$

(*d*) $K_1 > K_2 > K_3$

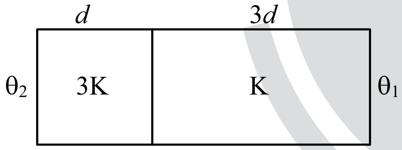
19. A heat source at $T = 10^3\ K$ is connected to another heat reservoir at $T = 10^2\ K$ by a copper slab which is 1 m thick. Given that the thermal conductivity of copper is $0.1\ \text{W K}^{-1}\text{m}^{-1}$, the energy flux through it in the steady state is: **[10 Jan, 2019 (Shift-I)]**

(*a*) $90\ \text{Wm}^{-2}$ (*b*) $120\ \text{Wm}^{-2}$
(*c*) $65\ \text{Wm}^{-2}$ (*d*) $200\ \text{Wm}^{-2}$

20. A cylinder of radius R is surrounded by a cylindrical shell of inner radius R and outer radius $2R$. The thermal conductivity of the material of the inner cylinder is K_1 and that of the outer cylinder is K_2. Assuming no loss of heat, the effective thermal conductivity of the system for heat flowing along the length of the cylinder is: **[12 Jan, 2019 (Shift-I)]**

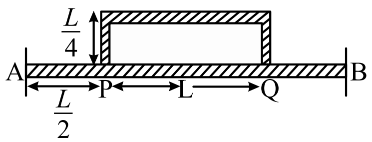
(*a*) $\dfrac{K_1 + K_2}{2}$ (*b*) $K_1 + K_2$
(*c*) $\dfrac{2K_1 + 3K_2}{5}$ (*d*) $\dfrac{K_1 + 3K_2}{4}$

21. Two materials having coefficients of thermal conductivity '$3K$' and 'K' and thickness 'd' and '$3d$', respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are 'θ_2' and 'θ_1' respectively, ($\theta_2 > \theta_1$). The temperature at the interface is: **[9 April, 2019 (Shift-II)]**



(*a*) $\dfrac{\theta_2 + \theta_1}{2}$ (*b*) $\dfrac{\theta_1}{10} + \dfrac{9\theta_2}{10}$
(*c*) $\dfrac{\theta_1}{3} + \dfrac{2\theta_2}{3}$ (*d*) $\dfrac{\theta_1}{6} + \dfrac{5\theta_2}{6}$

22. Temperature difference of 120°C is maintained between two ends of a uniform rod AB of length 2L. Another bent rod PQ, of same cross-section as AB and length $\dfrac{3L}{2}$, is connected across AB (See figure). In steady state, temperature difference between P and Q will be close to **[9 Jan, 2019 (Shift-I)]**



(*a*) 45°C (*b*) 75°C
(*c*) 60°C (*d*) 35°C

23. Match the temperature of a black body given in List-I with an appropriate statement in List-II, and choose the correct option. **[JEE Adv, 2023]**

[Given: Wien's constant as 2.9×10^{-3} m-K and $\dfrac{hc}{e} = 1.24 \times 10^{-6}$ V-m]

List-I		List-II	
(*p*)	2000 K	(*i*)	The radiation at peak wavelength can lead to emission of photoelectrons from a metal of work function 4 eV
(*q*)	3000 K	(*ii*)	The radiation at peak wavelength is visible to human eye.
(*r*)	5000 K	(*iii*)	The radiation at peak emission wavelength will result in the widest central maximum of a single slit diffraction.
(*s*)	10000 K	(*iv*)	The power emitted per unit area is 1/16 of that emitted by a black body at temperature 6000 K.
		(*v*)	The radiation at peak emission wavelength can be used to image human bones.

(*a*) $p \to (iii), q \to (v), r \to (ii), s \to (iii)$
(*b*) $p \to (iii), q \to (ii), r \to (iv), s \to (i)$
(*c*) $p \to (iii), q \to (iv), r \to (ii), s \to (i)$
(*d*) $p \to (i), q \to (ii), r \to (v), s \to (iii)$

Integer Type Questions

24. A unit scale is to be prepared whose length does not change with temperature and remains 20cm, using a bimetallic strip made of brass and iron each of different length. The length of both components would change in such a way that difference between their length remains constant. If length of brass is 40 cm and length of iron will be ________ cm. **[25 July, 2022 (Shift-I)]**

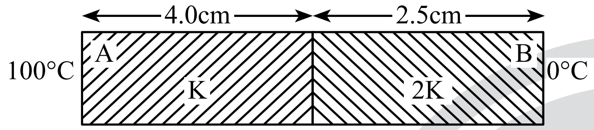
($\alpha_{iron} = 1.2 \times 10^{-5}\text{K}^{-1}$ and $\alpha_{brass} = 1.8 \times 10^{-5}\text{K}^{-1}$).

25. A steel rod with $y = 2.0 \times 10^{11}\ \text{Nm}^{-2}$ and $\alpha = 10^{-5}\ °\text{C}^{-1}$ of length 4 m and area of cross-section $10\ \text{cm}^2$ is heated from 0°C to 400°C without being allowed to extend. The tension produced in the rod is $x \times 10^5$ N where the value of x is __________. **[1 Sep, 2021 (Shift-II)]**

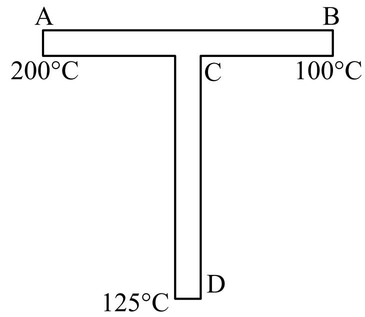
26. A non-isotropic solid metal cube has coefficients of linear expansion as: 5×10^{-5} /°C along the x-axis and 5×10^{-6} /°C along the y and the z-axis. If the coefficient of volume expansion of the solid is $C \times 10^{-6}$ /°C then the value of C is **[7 Jan, 2020 (Shift-I)]**

27. A bakelite beaker has volume capacity of 500 cc at 30°C. When it is partially filled with V_m volume (at 30°C) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{beaker} = 6 \times 10^{-6}\ °C^{-1}$ and $\gamma_{mercury} = 1.5 \times 10^{-4}\ °C^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to ___________. **[3 Sep, 2020 (Shift-I)]**

28. As per the given figure, two plates A and B of thermal conductivity K and $2K$ are joined together of form a compound plate. The thickness of plates are 4.0 cm and 2.5 cm respectively and the area of cross-section is 120 cm^2 for each plate. The equivalent thermal conductivity of the compound plate is $\left(1+\frac{5}{\alpha}\right)K$, then the value of α will be _______ . **[29 June, 2022 (Shift-I)]**



29. A rod CD of thermal resistance 10.0 kW^{-1} is joined at the middle of an identical rod AB as shown in figure, The end A, B and D are maintained at 200°C, 100°C and 125°C respectively. The heat current in CD is P watt. The value of P is **[27 Aug, 2021 (Shift-I)]**



ANSWER KEY

1. (*d*)	**2.** (*c*)	**3.** (*d*)	**4.** (*a*)	**5.** (*a*)	**6.** (*c*)	**7.** (*b*)	**8.** (*a*)	**9.** (*c*)	**10.** (*c*)
11. (*a*)	**12.** (*d*)	**13.** (*c*)	**14.** (*d*)	**15.** (*b*)	**16.** (*d*)	**17.** (*c*)	**18.** (*a*)	**19.** (*a*)	**20.** (*d*)
21. (*b*)	**22.** (*a*)	**23.** (*c*)	**24.** [60]	**25.** [8]	**26.** [60]	**27.** [20.00]	**28.** [21]	**29.** [2]	

EXPLANATIONS

1. (*d*) $\Delta\ell = 6.241 - 6.230 = 0.011$ cm

$\Delta\ell = \ell.\alpha\Delta\theta$

$0.011 = 6.230 \times 1.4 \times 10^{-5}(\theta - 27)$

$\theta \approx 153.11$ nearest is 152.7°C.

2. (*c*) As coefficient of thermal expansion of A is greater than B $(\alpha) = (\alpha)_A > (\alpha)_B$. So, with falling in temperature, A will shrink more. As a result, strip will bend towards the left.

3. (*d*) $(L_1 + L_2)\alpha_{eq} \times \Delta T = L_1\alpha_1\Delta T + L_1\alpha_2\Delta T$

$$\Rightarrow \alpha_{eq} = \frac{L_1\alpha_1 + L_2\alpha_2}{(L_1 + L_2)}$$

4. (*a*) Change in length due to rise in temperature $\Delta l = l\alpha\Delta T$

$\because \Delta l_1 = \Delta l_2$

$l\alpha_1\Delta T_1 = l\alpha_2\Delta T_2$

$$\frac{\alpha_1}{\alpha_2} = \frac{\Delta T_1}{\Delta T_2}$$

$$\frac{4}{3} = \frac{T-30}{180-30}$$

$T = 230°\text{C}$

5. (*a*) Given, heat energy $\Delta Q = 184 \times 10^3$

Energy Q, required to raise temperature of ice from –120°C to 0°C

$Q_1 = ms\Delta T = 0.600 \times 2222.3 \times 12 = 16000.56\ J$

Remaining heat $\Delta Q_1 = 184000 - 16000.56$

$= 167999.44$ J

For the ice to melt at 0°C

$\Delta Q_2 = mL = 0.600 \times 336000 = 201600\text{J}$ needed

$\therefore$ Total ice is not melted

Amount of ice melted

$167999.44 = \text{mass of ice} \times 336000 = 0.4999$ kg

$\therefore$ mass of water = 0.4999 kg

Mass of ice = 0.1001

$$\therefore \text{Ratio } = \frac{0.1001}{0.4999} \approx 1:5$$

6. (*c*) Heat given by block to get 0°C temperature

$\Delta Q = 5 \times (0.39 \times 10^3) \times (500 - 0) = 975 \times 10^3\text{J}$

Heat absorbed by ice to melt m kg mass

$m \times (335 \times 10^3) = 975 \times 10^3$

$$m = \frac{975}{335} = 2.910\text{kg}$$

7. (*b*) According to the question,

Heat gained by liquid x = Heat lost by liquid y

$ms_x(16 - 10) = ms_y(20 - 16)$

$$\Rightarrow \frac{s_x}{s_y} = \frac{2}{3} \quad \text{... } (i)$$

$ms_y(26 - 20) = ms_z(30 - 26)$

$$\Rightarrow \frac{s_y}{s_z} = \frac{2}{3} \quad \text{... } (ii)$$

From equation (*i*) and (*ii*)

$$\frac{s_x}{s_z} = \frac{s_x}{s_y} \times \frac{x_y}{s_z} = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$$

When liquid x and z are intermixed

$ms_x(T - 10) = ms_z(30 - T)$

$\Rightarrow 4(T - 10) = 9(30 - T)$ [from (*ii*)]

$\Rightarrow \text{T} = 23.84°\text{C}$

8. (*a*) Change in potential energy = Heat energy

$mgh = ms\Delta T$

$$\Rightarrow \Delta T = \frac{gh}{s} = \frac{10 \times 63}{4200} = 0.147°\,C$$

9. (*c*) Required energy for melting

$Q = MS\Delta T + ML$

$\Rightarrow 10^{-1} \times 2\times 10^3 \times 10 + 10^{-1} \times 3.33 \times 10^5$

$\Rightarrow 3.53 \times 10^4$ J

Thus, heat produce in wire is

$\Rightarrow H = I^2$ Rt

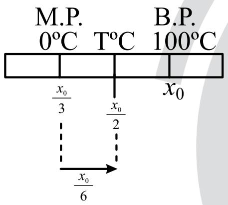
$$\Rightarrow Q = 3.53\times10^4 = \left(\frac{1}{2}\right)^2 \times(4\times10^3)\times t$$

$$t = \frac{3.53\times10^4\times4}{4\times10^3} = 35.3\,\text{sec}$$

10. (*c*) Here $\frac{1}{2}\left(\frac{1}{2}mv^2\right) = mS\Delta T$

$$\Rightarrow \Delta T = \frac{v^2}{45} = \frac{(210,00)}{4\times0.0\times4.2\times10^7} \simeq 84.5°C$$

11. (*a*) $t = \dfrac{x_t - x_0}{x_{100} - x_0}(100°\text{C})$



$$t = \frac{\frac{x_0}{2} - \frac{x_0}{3}}{x_0 - \frac{x_0}{3}}(100°\text{C}) = 25°\text{C}$$

12. (*d*) $m \times 0.5 \times 20 + (m - 20) \times 80 = 50 \times 1 \times 40$

$\Rightarrow 90m - 1600 = 2000$

$\Rightarrow 90m = 3600 \Rightarrow m = 40$ gm

13. (*c*) $100 \times S_A \times [100 - 90] = 50 \times S_B \times (90 - 75)$

$2S_A = 1.5\ S_B$

$S_A = \frac{3}{4}S_B$

Now, $100 \times S_A \times [100 - T] = 50 \times S_B\,(T - 50)$

$2 \times \left(\frac{3}{4}\right)(100 - T) = (T - 50)$

$300 - 3T = 2T - 100$

$400 = 5T$

$T = 80°\text{C}$

14. (*d*) $0.1 \times 400 \times (500 - T) = 0.5 \times 4200 \times (T - 30) + 800\,(T - 30)$

$\Rightarrow 40(500 - T) = (T - 30)(2100 + 800)$

$\Rightarrow 20000 - 40\,T = 2900\,T - 30 \times 2900$

$\Rightarrow 20000 + 30 \times 2900 = T(2940) \Rightarrow T = 36.4°\text{C}$

$$\frac{\Delta T}{T}\times100 = \frac{6.4}{30}\times100 \approx 20\%$$

15. (*b*) $\dfrac{dQ}{dt} = \dfrac{KA\Delta T}{\ell} = \dfrac{\Delta T}{R_{\text{thermal}}}$

Area = $2 \times [\ell b + bh + h\ell]$

$\Rightarrow A = 2 \times [0.6 \times 0.5 + 0.5 \times 0.2 + 0.2 \times 0.6] = 1.04\ \text{m}^2$

$$R_{\text{thermal}} = \frac{\ell}{KA} = \frac{1\times10^{-2}}{0.05\times1.04}$$

$$\therefore \frac{dQ}{dt} = \frac{40\times0.05\times1.04}{10^{-2}} = 2.08 \times 10^2\ \text{J/s}$$

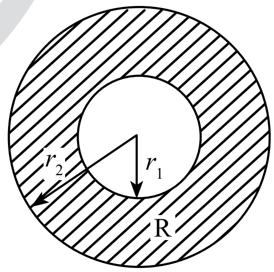
Now, let m kg/s be the rate of melting of ice

$\Rightarrow 2.08 \times 10^2$ J/S $= m \times 3.4 \times 10^5$

$\Rightarrow m = 61 \times 10^{-5}$ kg/s

16. (*d*) Thermal resistance of spherical shell $= \dfrac{dr}{4\pi K r_1 r_2}$

$$\text{Thermal current } \frac{\Delta T}{R} = \frac{(\theta_2 - \theta_1)}{\left(\frac{r_2 - r_1}{4\pi K r_1 r_2}\right)} = \frac{4\pi K r_1 r_2(\theta_2 - \theta_1)}{(r_2 - r_1)}$$



17. (*c*) Formula of thermal resistance, $R = \dfrac{l}{KA}$

$$R_{eq} = R_1 + R_2 = \frac{\ell}{K_1 A} + \frac{\ell}{K_2 A} = \frac{\ell}{A}\left(\frac{1}{K_1} + \frac{1}{K_2}\right) = \frac{2\ell}{KA}$$

$$\Rightarrow K = \frac{2K_1 K_2}{K_1 + K_2}$$

18. (*a*) $K_1\,(100 - 70) = K_2\,(50)$

$K_3\,(20) = K_2\,(50)$

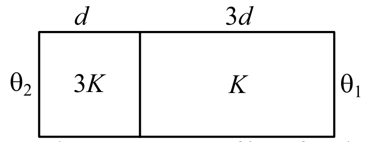
19. (*a*) $\frac{\Delta Q}{\Delta T}=\frac{kA}{\ell}(T_2-T_1) \Rightarrow \frac{1}{A}\left(\frac{\Delta Q}{\Delta T}\right)=\frac{k}{\ell}(T_2-T_1)$

20. (*d*) Equivalent thermal resistance, $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}$

$$\frac{K\pi(2R)^2}{L}=\frac{K_1\pi R^2}{L}+\frac{K_2\pi\left[(2R)^2-R^2\right]}{L}$$

$$\Rightarrow 4K=K_1+3K_2 \Rightarrow K=\frac{K_1+3K_2}{4}$$

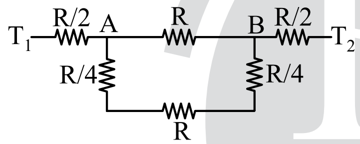
21. (*b*)



Let the temperature of interface be "θ" $i_1 = i_2$ (Steady state conduction)

$\frac{3KA(\theta_2-\theta)}{d}=\frac{KA(\theta-\theta_1)}{3d}$ $\theta=\frac{9\theta_2}{10}+\frac{\theta_1}{10}$

22. (*a*) $T_A-T_B=\frac{T_1-T_2}{\frac{8R}{5}}\times\frac{3R}{5}=\frac{3}{8}\times120=45$



23. (*c*) Weins displacement law, $\lambda_m T = b$(constant)

Temperature is minimum for *P*, therefore λ_m will be maximum and $\beta=\left(\frac{\lambda D}{d}\right)$ therefore, β will also be maximum.

For (*Q*), $T_Q = 3000$

$\therefore\ \lambda_m=\frac{b}{T_Q}=\frac{2.9\times10^{-3}}{30000}$ $\lambda_m=\frac{2.9}{3}\times10^{-6}$

$= 0.96 \times 10^{-6} = 966.6$ nm

Power emitted per unit area, $P_E = \delta T^4$

$P_{3000} \propto (3000)^4$

$P_{6000} \propto (6000)^4$

$\therefore \frac{P_{3000}}{P_{6000}}=\left(\frac{1}{2}\right)^4=\frac{1}{16}$

$P_{3000}=\frac{1}{16}P_{6000}$

$\Rightarrow$ For (*R*), $T_R = 5000$ K

$\lambda_m=\frac{2.9\times10^{-3}}{5\times10^3}=0.58\times10^{-6}=580$ nm

Hence, radiation is visible to human eyes.

$\Rightarrow$ For (*S*), $T_s = 10,000 \rightarrow$ maximum

So, (*c*) is wrong as it has minimum (λ_m)

24. [60] $\ell_B(1+\alpha_B\Delta T)-\ell_i(1+\alpha_i\Delta T)=\ell_B-\ell_i$

From $L = L_0(1+\alpha\Delta T) = L_0 + L_0\alpha\Delta T$

$\Delta T = (L_0)\alpha\Delta T$

$\therefore\ \Delta L_A = (L_0)_A\alpha_A\Delta T$ ...(*i*)

$\Delta L_B = L_{0B}\alpha_B\Delta T$ ...(*ii*)

from (*i*) and (*ii*)

$(L_0)_A\alpha_A = (L_0)_B\alpha_B$

$= (L_0)_A \times 1.2\times10^{-5} = 1.8\times10^{-5}\times40$

$$(L_0)_A=\frac{1.8\times10^{-5}\times40}{1.2\times10^{-5}}=\frac{3\times40}{2}=60\,\text{cm}$$

25. [8] Given: $y = 2\times10^{11}$ Nm^{-2}

$\alpha = 10^{-50}$ C^{-1}

Force = $Ay\alpha\Delta T$

Force = $(10\times10^{-4})\times(2\times10^{11})\times10^{-5}\times400$

$F = 8\times10^5$ N

$x\times10^5 = 8\times10^5$

Thus, the value of is $x = 8$

26. [60] $V = 2\alpha_2 + \alpha_1$
$= 10\times10^{-6} + 5\times10^{-5}$
$= 60\times10^{-6}/°C$

27. [20.00]

$V_0 = 500$ cc

$V_b = V_0 + V_0\gamma_{\text{beaker}}\Delta T$

And for Mercury

$V_m' = V_m + V_m\gamma_m\Delta T$

Unfilled volume $(V_0 - V_m) = (V_b - V'_m)$

$\Rightarrow V_0\gamma_{\text{beaker}} = V_m - \gamma_M$

$\therefore\ V_m=\frac{500\times6\times10^{-6}}{1.5\times10^{-4}} \Rightarrow V_m = 20$ cc

28. [21] It is given that,

$L_1 = 4$cm, $L_2 = 2.5$cm, $K_1 = K_2 = 2K$

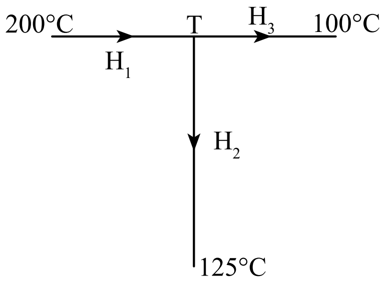
$$K_{eq}=\frac{L_1+L_2}{\frac{L_1}{K_1}+\frac{L_2}{K_2}}=\frac{4+2.5}{\frac{4}{K}+\frac{2.5}{2K}}$$

$$K_{eq}=\frac{6.5}{\frac{8+2.5}{2K}}=\frac{6.5}{10.5}\times2K=\frac{13\times2}{21}K=\left(1+\frac{5}{51}\right)K$$

By equating the above equation with $k_{eq} = \left(1+\frac{5}{\alpha}\right)$

We get, $\alpha = 21$

29. [2] At puint T



$$\Rightarrow H_1 = H_2 + H_3$$

$$\Rightarrow \frac{200-T}{5} = \frac{T-125}{10} + \frac{T-100}{5}$$

$$\Rightarrow T = 145^\circ C \Rightarrow H_2 = \frac{145-125}{10} = 2W$$